

Lab 01 - Linear Models, Cross Validation

[Your Name Here]

Due 19 September at 11pm

Instructions

Brief description

In this lab, we will work on the `prostate` cancer data set. The (modified) data set has observations and variables; the variable `lpsa` in the 9th column is the response while the rest are potential predictors. The data set is available in the `{ElemStatLearn}` package and a description can be found at <http://statweb.stanford.edu/~tibs/ElemStatLearn/datasets/prostate.info.txt>

Questions

Load the data into R and remove the `train` column.

```
data("prostate", package = "ElemStatLearn")
library(tidyverse)
prostate <- prostate |> select(-train)
```

You can view the first few observations by typing `head(prostate)`. In this lab, we are interested in two linear models: a. The **[full model]** with all 8 predictors, and a. The **[reduced model]** with the 3 predictors most correlated with the response `lpsa`.

Q1

Which 3 predictors are the most correlated with `lpsa`?

```
# Compute correlations with lpsa
cors <- cor(prostate)[, "lpsa"]

# Sort by absolute correlation (strongest first)
sort(abs(cors), decreasing = TRUE)
```

```
##      lpsa      lcavol      svi      lcp      lweight      pgg45      gleason      lbph
## 1.0000000 0.7344603 0.5662182 0.5488132 0.4333194 0.4223159 0.3689868 0.1798094
##      age
## 0.1695928
```

```
prostate %>%
  select(-lpsa) %>% # drop response so only predictors remain
  summarise(across(everything(),
                    ~ cor(.x, prostate$lpsa))) %>% # compute correlation with lpsa
  pivot_longer(everything(),
               names_to = "predictor",
               values_to = "correlation") %>%
  mutate(abs_corr = abs(correlation)) %>%
  arrange(desc(abs_corr)) %>%
  slice_head(n = 3)
```

```
## # A tibble: 3 x 3
##   predictor correlation abs_corr
##   <chr>         <dbl>     <dbl>
## 1 lcavol       0.734     0.734
## 2 svi          0.566     0.566
## 3 lcp          0.549     0.549
```

Q2

Without fitting the linear models, can you tell which model fits the data better in the sense of having a smaller residual sum of squares? Why?

Your answer

Next, we estimate the prediction error using the test set approach as well as cross validation. Enter the following command in R to generate a random sequence of group indices (group 1, 2 or 3) for the observations.

Note: Using another random seed, or not using one, or running the `sample` command more than once will give you a different sequence than the one we will use for marking this worksheet — make sure you get this right.

```
set.seed(400)
gpind <- c(sample(rep(1:3, each = 32)), 1)
```

The first 6 elements of `gpind` should read 1, 3, 3, 3, 1, 1.

Q3

Fit the full and reduced models using observations [not] in group 1, and then obtain predictions for those in group 1. What are the mean squared prediction errors for the two models?

Hint: The reduced model uses the 3 predictors most correlated with the response in the training set. Not necessarily the same as those above that used *all* the data.

```

# # Training set = not in group 1
# train <- prostate[gpind != 1, ]
#
# # Test set = in group 1
# test  <- prostate[gpind == 1, ]

# Add group index as a column
prostate2 <- prostate %>%
  mutate(group = gpind)

# Training = not in group 1
train <- prostate2 %>% filter(group != 1)

# Test = in group 1
test  <- prostate2 %>% filter(group == 1)

# --- Full model (all 8 predictors, exclude 'group') ---
fit_full <- lm(lpsa ~ . - group, data = train)

# --- Reduced model (lcavol, svi, lcp only) ---
fit_reduced <- lm(lpsa ~ lcavol + svi + lcp, data = train)

# --- Predictions on test set (group 1) ---
pred_full <- predict(fit_full, newdata = test)
pred_reduced <- predict(fit_reduced, newdata = test)

# --- Mean Squared Prediction Error ---
mspe_full <- mean((test$lpsa - pred_full)^2)
mspe_reduced <- mean((test$lpsa - pred_reduced)^2)

mspe_full

## [1] 0.563016

mspe_reduced

## [1] 0.6973839

```

Q4

Does your result in **Q3** contradict with that in **Q2**, and why?

Your answer

Q5

Repeat the procedure in **Q3** for groups 2 and 3, so as to perform a 3-fold cross validation. What are the overall mean squared prediction errors for the two models?

```

# Define reduced model predictors (from training set analysis earlier)
reduced_predictors <- c("lcavol", "svi", "lcp")

# Storage for MSPEs
mspe_full <- c()
mspe_reduced <- c()

# Loop over groups 1, 2, 3
for (g in 1:3) {
  # Split into train/test
  train <- prostate2 %>% filter(group != g)
  test  <- prostate2 %>% filter(group == g)

  # Fit models
  fit_full <- lm(lpsa ~ . - group, data = train)
  fit_reduced <- lm(
    as.formula(paste("lpsa ~", paste(reduced_predictors, collapse = " + "))),
    data = train
  )

  # Predictions
  pred_full <- predict(fit_full, newdata = test)
  pred_reduced <- predict(fit_reduced, newdata = test)

  # Compute MSPE for this fold
  mspe_full[g] <- mean((test$lpsa - pred_full)^2)
  mspe_reduced[g] <- mean((test$lpsa - pred_reduced)^2)
}

# Overall mean MSPEs across 3 folds
mean_mspe_full <- mean(mspe_full)
mean_mspe_reduced <- mean(mspe_reduced)

mean_mspe_full

## [1] 0.5405286
mean_mspe_reduced

## [1] 0.591308

```

Q6

Based on these results, which model has a better predictive ability? Suggest one advantage of the K -fold cross validation ($K > 1$) over the leave-one-out cross validation.

Your answer